

U6 Recommended standard battery charge current is 100mA, using ISET R=8.9kOhm (From vendor's Amazon page). Resistor for ILIM should be 1.18kOhm for a current input of 1.3A

Connect a 1100-Ω to 8-kΩ resistor from ILIM to VSS to program the maximum input current (EN2=1, EN1=0). The input current includes the system load and the battery charge current. Leaving ILIM unconnected disables all charging. U6

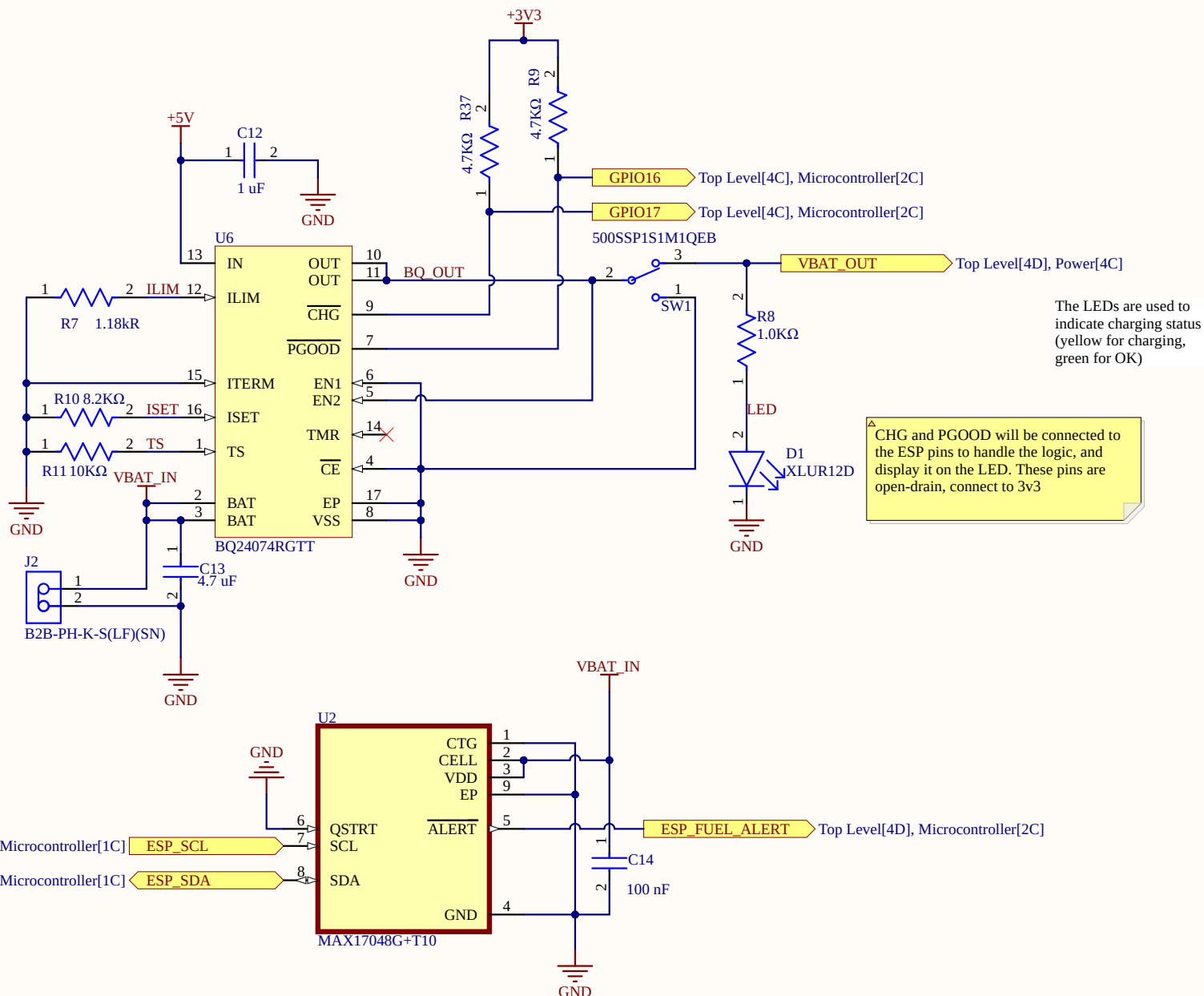
Connect a 0-Ω to 15-kΩ resistor from ITERM to VSS to program the termination current. Leave ITERM unconnected to set the termination current to the default 10% termination threshold.

J2 has pitch = 2.0mm, matching the specs on the battery purchased. It has current rating of 2A, which will be suitable for our application

The ISET pin sets the charge current via a resistor to GND. Formula: $V_{ISET} = I_{CHARGE} / 400 \times R_{ISET}$

Top Level[4C], HeartrateMonitor[2C], Microcontroller[1C] ESP_SCL

Top Level[4C], HeartrateMonitor[2C], Microcontroller[1C] ESP_SDA



The LEDs are used to indicate charging status (yellow for charging, green for OK)

CHG and PGOOD will be connected to the ESP pins to handle the logic, and display it on the LED. These pins are open-drain, connect to 3v3

Title			Battery Management System		
Size	Number		Revision		
A	LUM-2026-SCH-005		v1.0		
Date:	5/22/2026		Sheet 5 of	9	
File:	C:\Users\...\BMS.SchDoc		Drawn By:	Joe Nguyen	